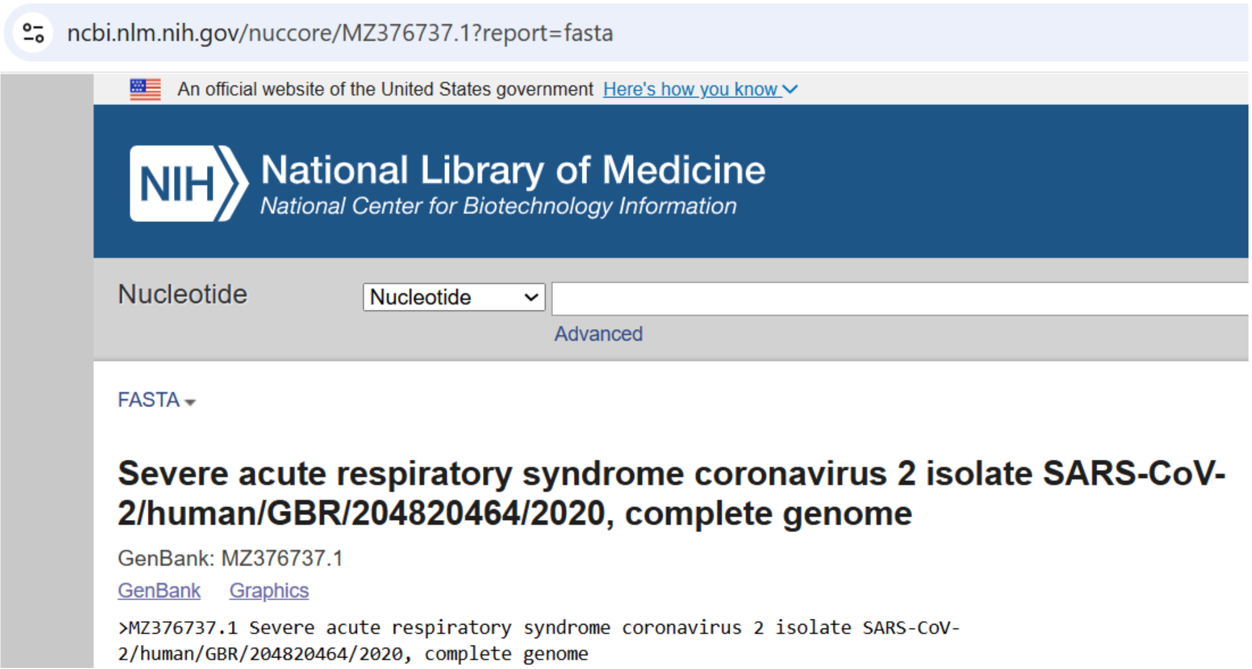
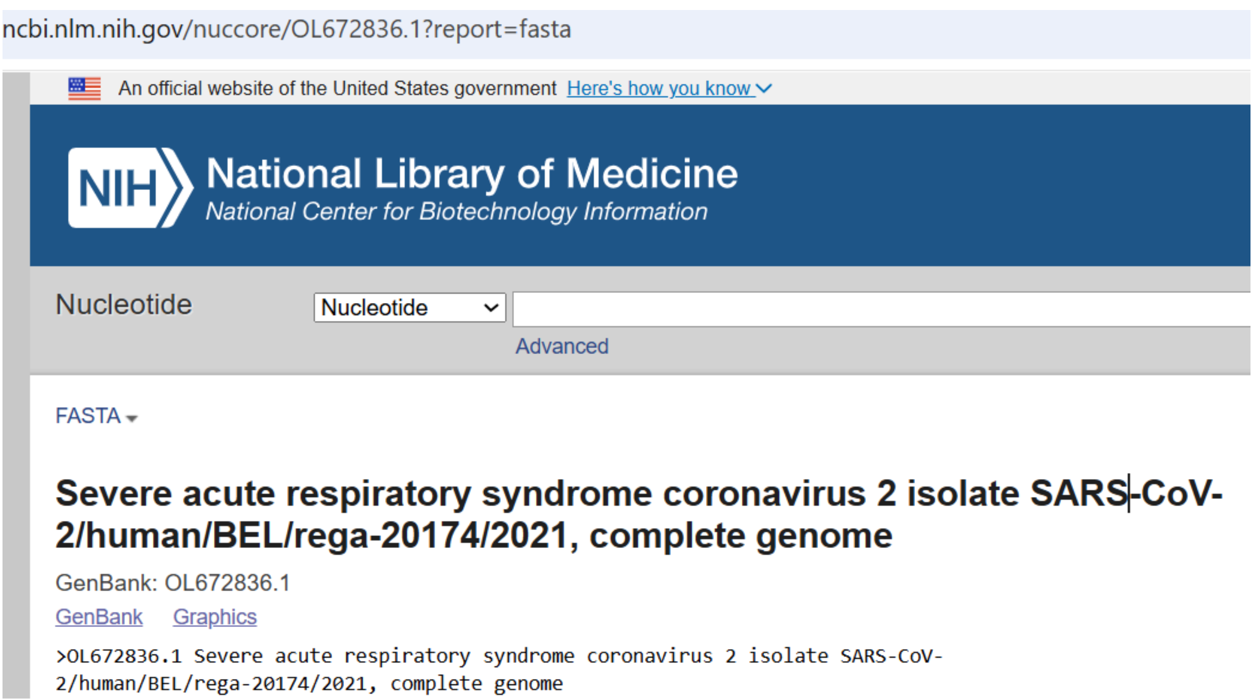
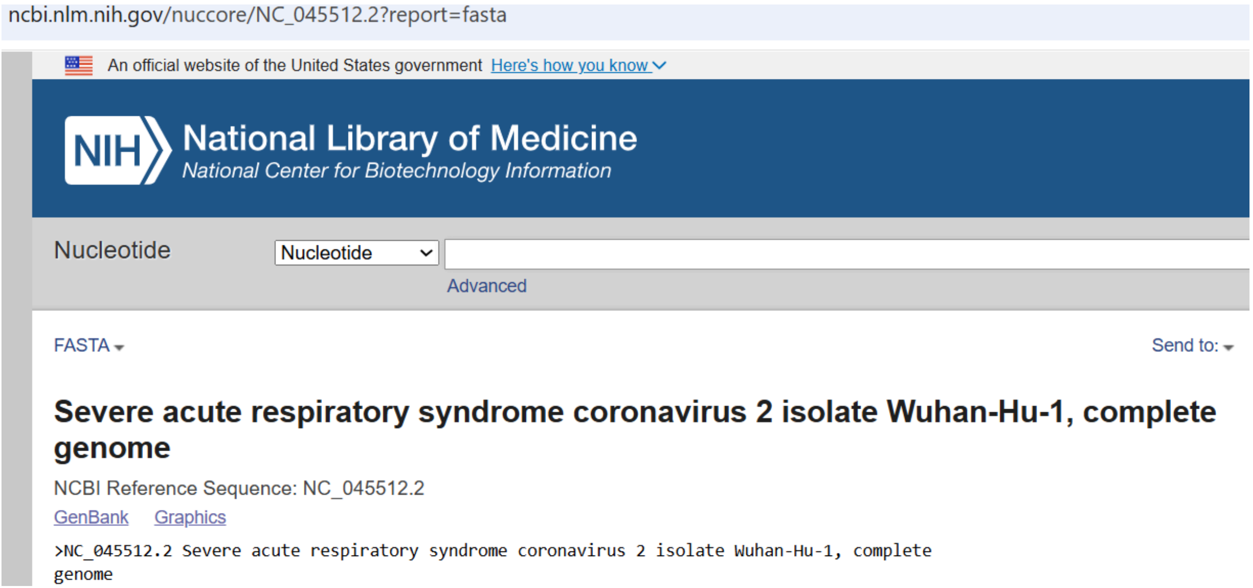
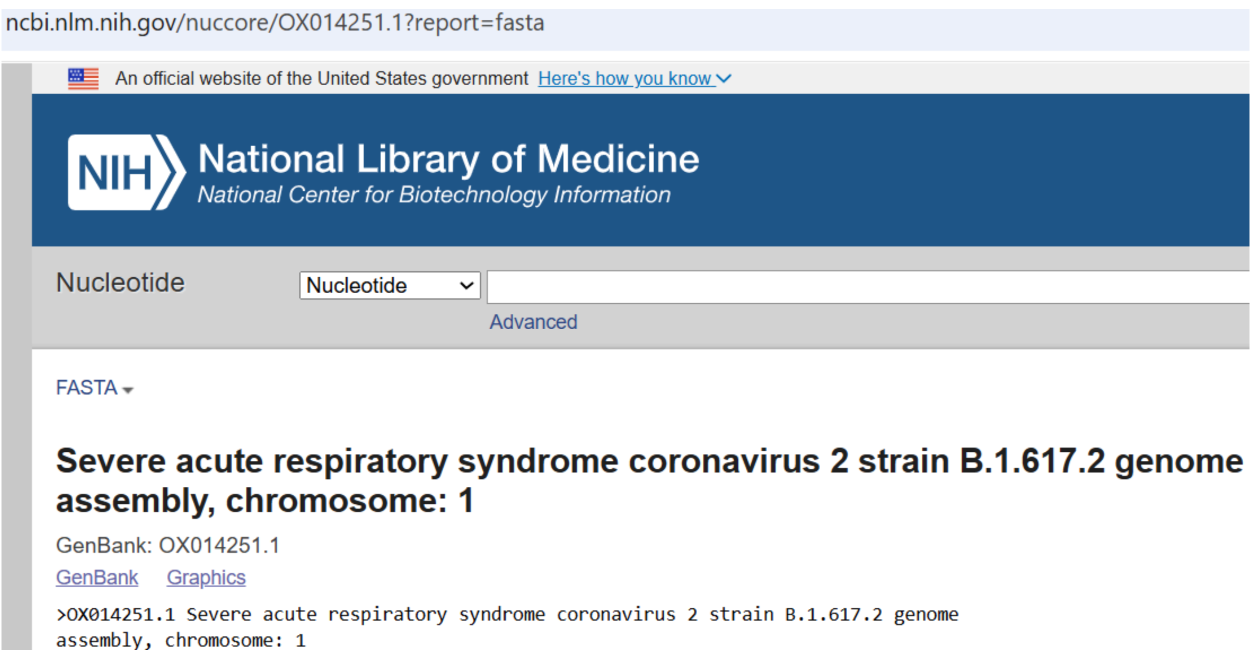
1. Найти от 3 до 5 последовательностей S-белка SARS-CoV-2 в открытых источниках. Добавить к ним какую-нибудь из последовательностей, рассмотренных на семинаре. Построить для них филогенетическое дерево, сделать выводы.

*<u>Решение:</u>*






**Получившийся fasta-файл:**

>NC_045512.2 Severe acute respiratory syndrome coronavirus 2 isolate Wuhan-Hu-1
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>OX014251.1 Severe acute respiratory syndrome coronavirus 2 strain B.1.617.2 genome assembly, chromosome: 1
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>OL672836.1 Severe acute respiratory syndrome coronavirus 2 isolate
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>MZ376737.1 Severe acute respiratory syndrome coronavirus 2 isolate
SARS-CoV-2/human/GBR/204820464/2020
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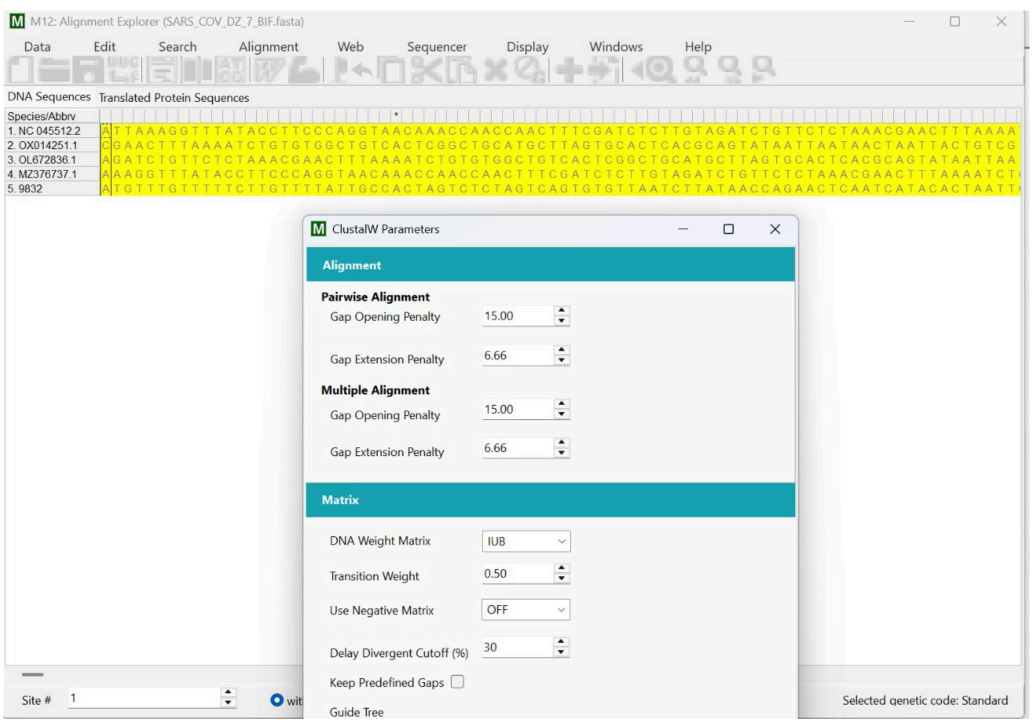
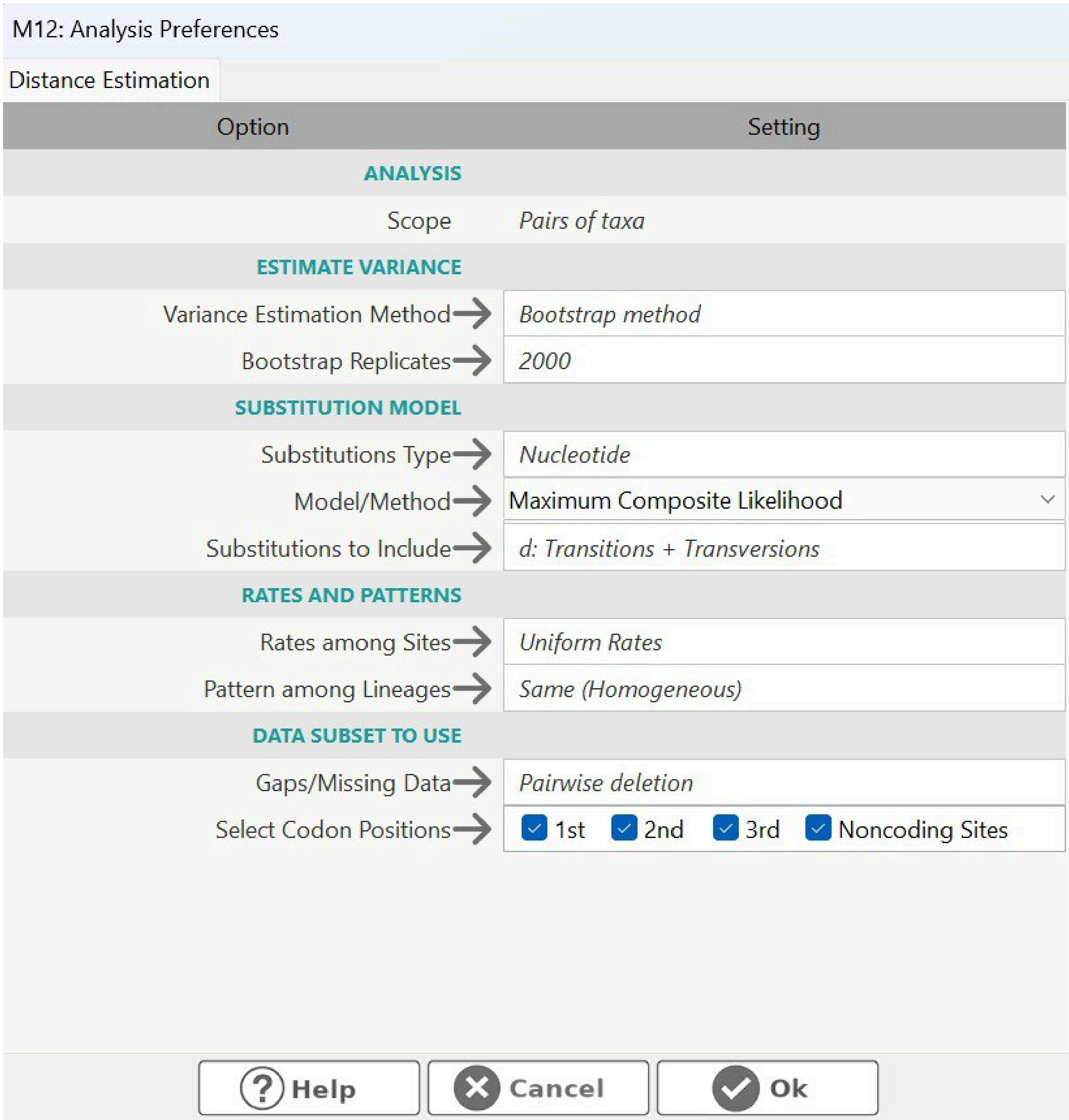
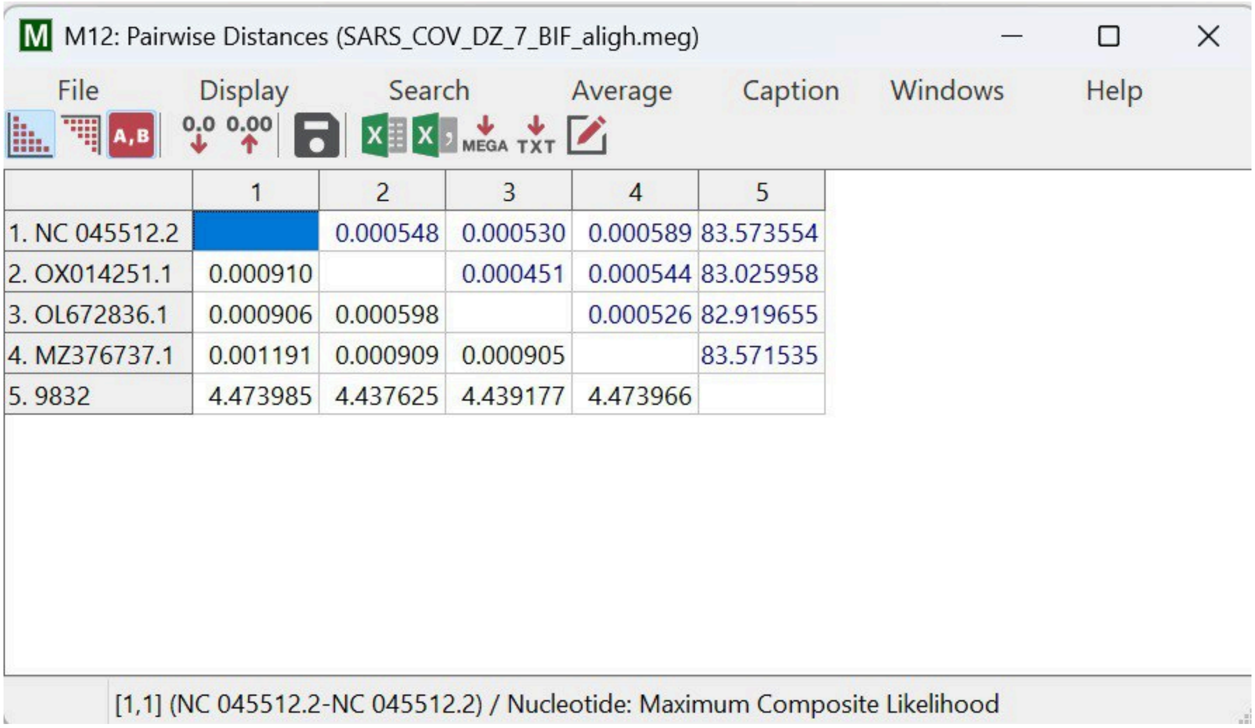
**Далее мы выровнили последовательности в MEGA с помощью алгоритма Clustalw, импортировали результат в mega-файл и построили филогенетическое дерево.**




M12: Analysis Preferences
Distance Estimation
Option
Setting
ANALYSIS
Scope
Pairs of taxa
ESTIMATE VARIANCE
Variance Estimation Method
Bootstrap method
Bootstrap Replicates
2000
SUBSTITUTION MODEL
Substitutions Type
Nucleotide
Model/Method
Maximum Composite Likelihood
Substitutions to Include
d: Transitions + Transversions
RATES AND PATTERNS
Rates among Sites
Uniform Rates
Pattern among Lineages
Same (Homogeneous)
DATA SUBSET TO USE
Gaps/Missing Data
Pairwise deletion
Select Codon Positions
1st
2nd
3rd
Noncoding Sites
Help
Cancel
Ok



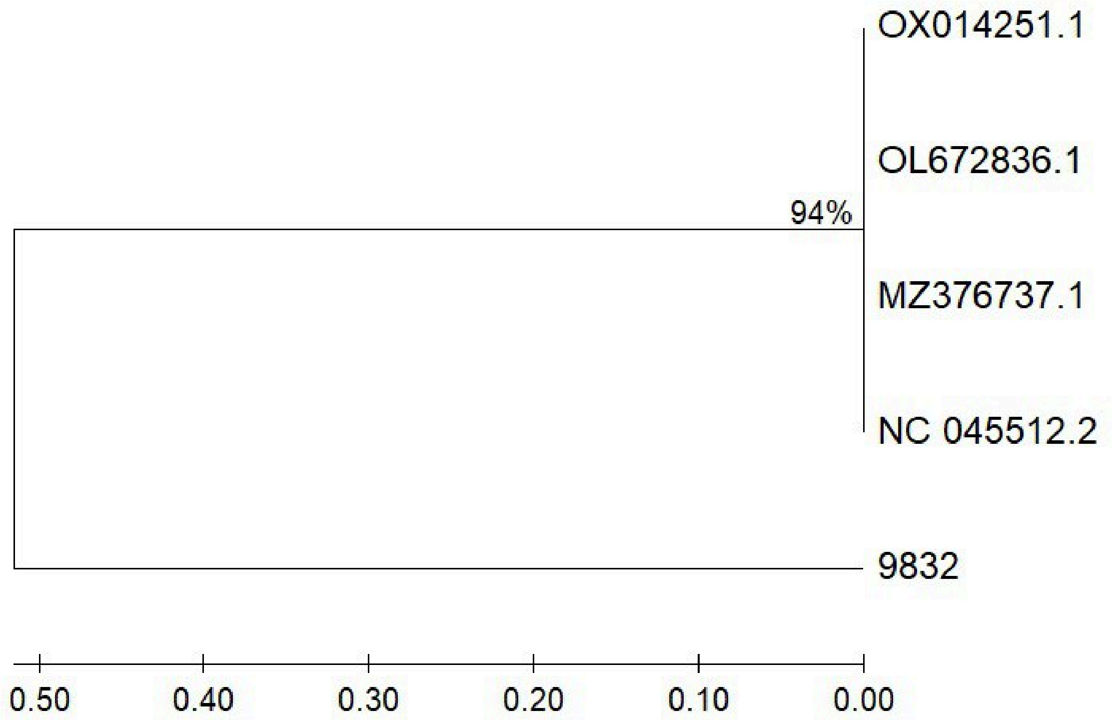

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1. NC 045512.2 | | 0.000548 | 0.000530 | 0.000589 | 83.573554 |
| 2. OX014251.1 | 0.000910 | | 0.000451 | 0.000544 | 83.025958 |
| 3. OL672836.1 | 0.000906 | 0.000598 | | 0.000526 | 82.919655 |
| 4. MZ376737.1 | 0.001191 | 0.000909 | 0.000905 | | 83.571535 |
| 5. 9832 | 4.473985 | 4.437625 | 4.439177 | 4.473966 | |

По получившейся матрице расстояний строим филогенетическое дерево:

M12: Analysis Preferences

Phylogeny Reconstruction

| Option | Setting |
|---|---|
| **ANALYSIS** | |
| Scope | *All Selected Taxa* |
| Statistical Method → | *UPGMA* |
| **PHYLOGENY TEST** | |
| Test of Phylogeny → | *Bootstrap method* |
| Bootstrap Replicates → | *2000* |
| **SUBSTITUTION MODEL** | |
| Substitutions Type → | *Nucleotide* |
| Model/Method → | *Kimura 2-parameter model* |
| Substitutions to Include → | *d: Transitions + Transversions* |
| **RATES AND PATTERNS** | |
| Rates among Sites → | *Uniform Rates* |
| Pattern among Lineages → | *Same (Homogeneous)* |
| **DATA SUBSET TO USE** | |
| Gaps/Missing Data → | *Pairwise deletion* |
| Select Codon Positions → | ☑ 1st ☑ 2nd ☑ 3rd ☑ Noncoding Sites |
| **SYSTEM RESOURCE USAGE** | |
| Number of Threads → | *3* |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MZ376737.1
NC 045512.2
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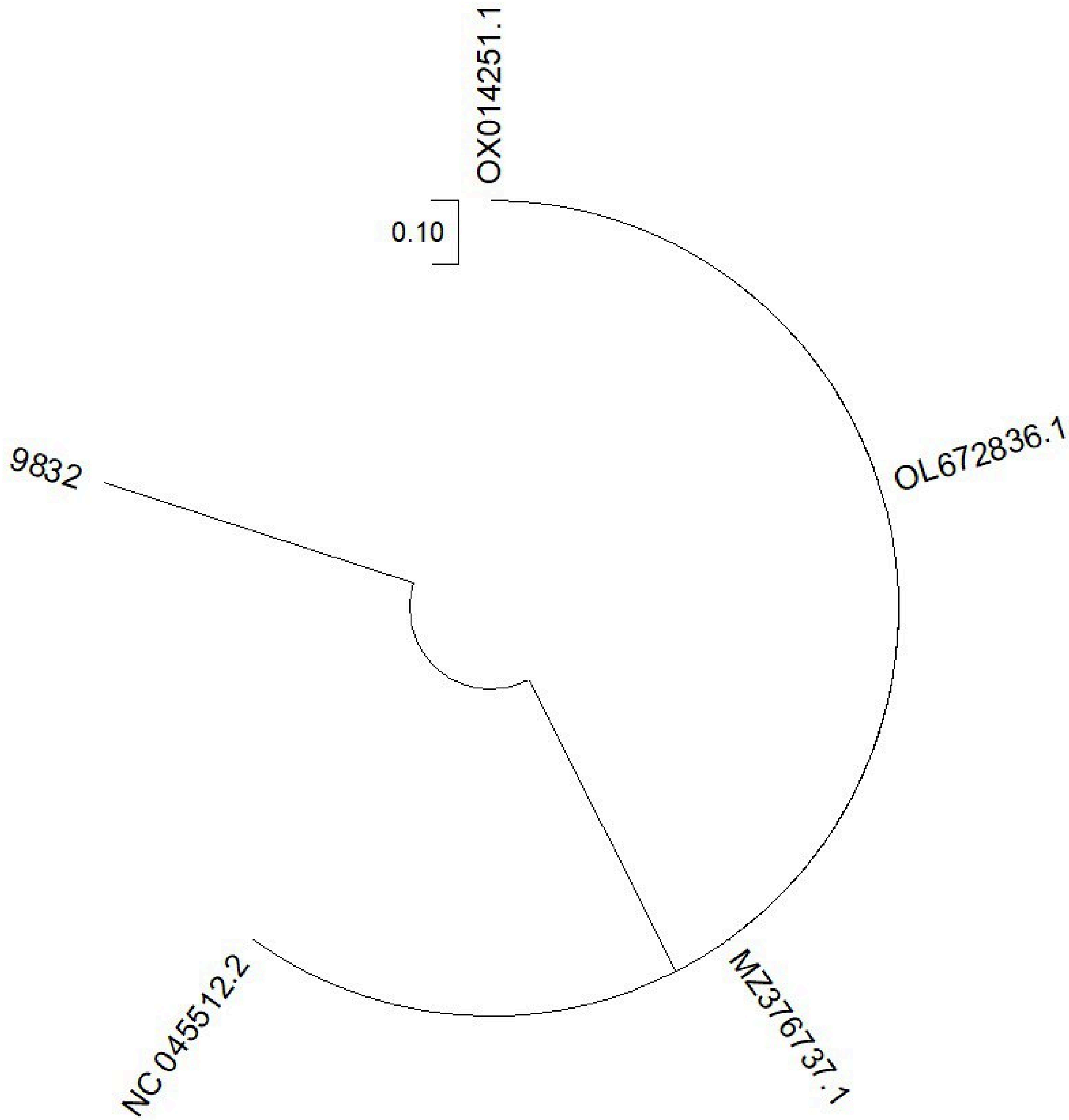
0.40
0.30
0.20
0.10
0.00




Выводы:

1. Последовательности NO. 045512.2 и 9822 образуют близкую кладу, что указывает на их высокое сходство и возможное происхождение от общего предка.
2. Последовательность 0.10 демонстрирует более отдалённое родство с остальными, что может свидетельствовать о её уникальных мутациях или отдельной эволюционной линии.
3. Группа OX014251.1 и OL672836.1 также близкородственна, но дистанцирована от первых двух клад, что подчёркивает разнообразие вариантов вируса.

4. Дерево отражает наличие как минимум трёх основных ветвей, что может быть связано с разными географическими или временными источниками образцов.

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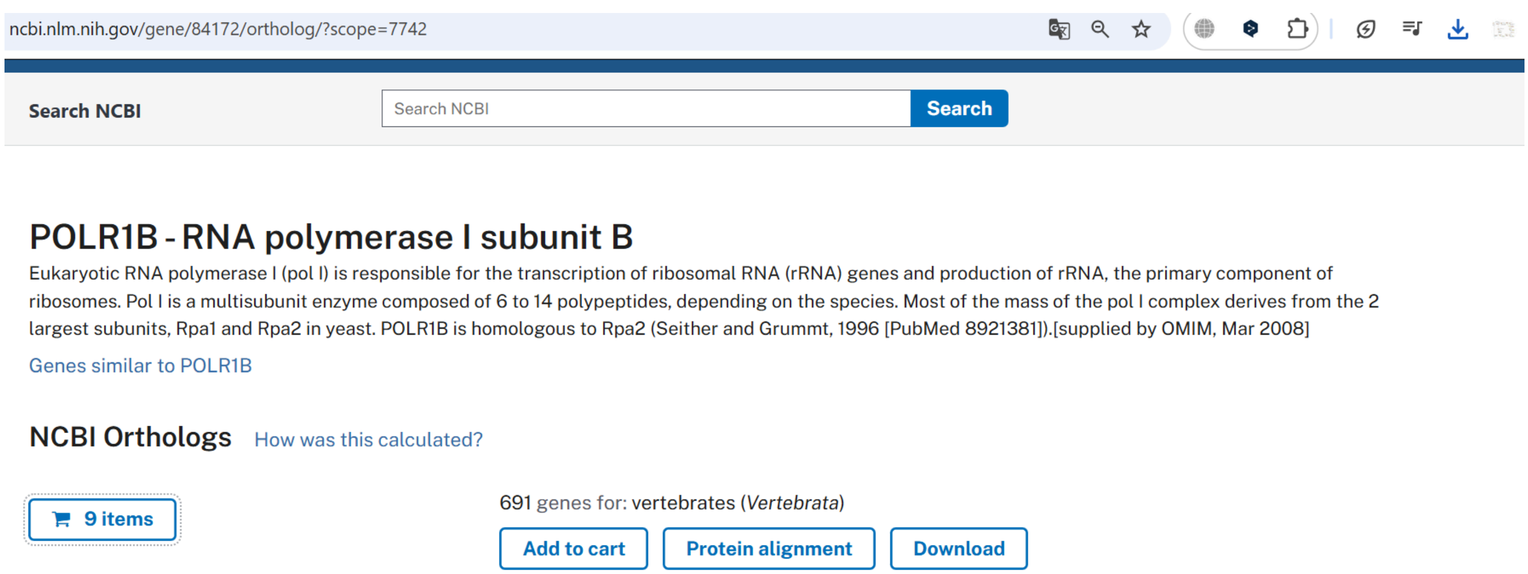
2. Найти в базе данных ортологов последовательности (от 4 до 10) на ваш вкус и построить для них филогенетическое дерево, также сделать выводы. Как найти ортологи: прочитайте форум (https://www.biostars.org/p/493467/) и воспользуйтесь туториалом (https://www.ncbi.xyz/datasets/docs/v2/tutorials/download-ortholog-dataset/). Также можно просто зайти на NCBI (https://www.ncbi.nlm.nih.gov/search/all/) и ввести название гена, ортологи которого хотите найти. В результатах поиска нажмите "Orthologs", после чего вам выведутся все ортологи данного гена.

*Решение:*

Опять надо вспомнить(хотя я никак не могу этого запомнить), что ортологи — это гены в разных организмах, которые произошли от общего предка в результате видообразования…

Искали ген rpoB.



**Cart**

Protein alignment | Download | Remove all

9 selected

| Species | Gene | RNA | Protein | Remove |
|---|---|---|---|---|
| *Mus musculus*<br>house mouse | Polr1b<br>polymerase (RNA) I polypeptide B | 3 | 2 | |
| *Bos taurus*<br>domestic cattle | POLR1B<br>RNA polymerase I subunit B | 2 | 2 | |
| *Lates calcarifer*<br>barramundi perch | polr1b<br>RNA polymerase I subunit B | 1 | 1 | |
| *Cricetulus griseus*<br>Chinese hamster | Polr1b<br>RNA polymerase I subunit B | 4 | 4 | |
| *Oryzias melastigma*<br>Indian medaka | polr1b<br>RNA polymerase I subunit B | 1 | 1 | |
| *Seriola dumerili*<br>greater amberjack | polr1b<br>RNA polymerase I subunit B | 1 | 1 | |
| *Nannospalax galili*<br>Upper Galilee mountains blind mole rat | Polr1b<br>RNA polymerase I subunit B | 1 | 1 | |
| *Astyanax mexicanus*<br>Mexican tetra | polr1b<br>RNA polymerase I subunit B | 1 | 1 | |
| *Oryzias latipes*<br>Japanese medaka | polr1b<br>RNA polymerase I subunit B | 1 | 1 | |

# Наш fasta-файл:

```
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>XM_003497791.4 PREDICTED: Cricetulus griseus RNA polymerase I subunit B (Polr1b), transcript variant X1, mRNA
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>XM_004076892.4 PREDICTED: Oryzias latipes RNA polymerase I subunit B (polr1b), mRNA
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>XM_007258693.4 PREDICTED: Astyanax mexicanus RNA polymerase I subunit B (polr1b), mRNA
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>XM_008840365.3 PREDICTED: Nannospalax galili RNA polymerase I subunit B (Polr1b), mRNA
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>XM_018681023.2 PREDICTED: Lates calcarifer RNA polymerase I subunit B (polr1b), mRNA
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>XM_022743766.1 PREDICTED: Seriola dumerili RNA polymerase I subunit B (polr1b), mRNA
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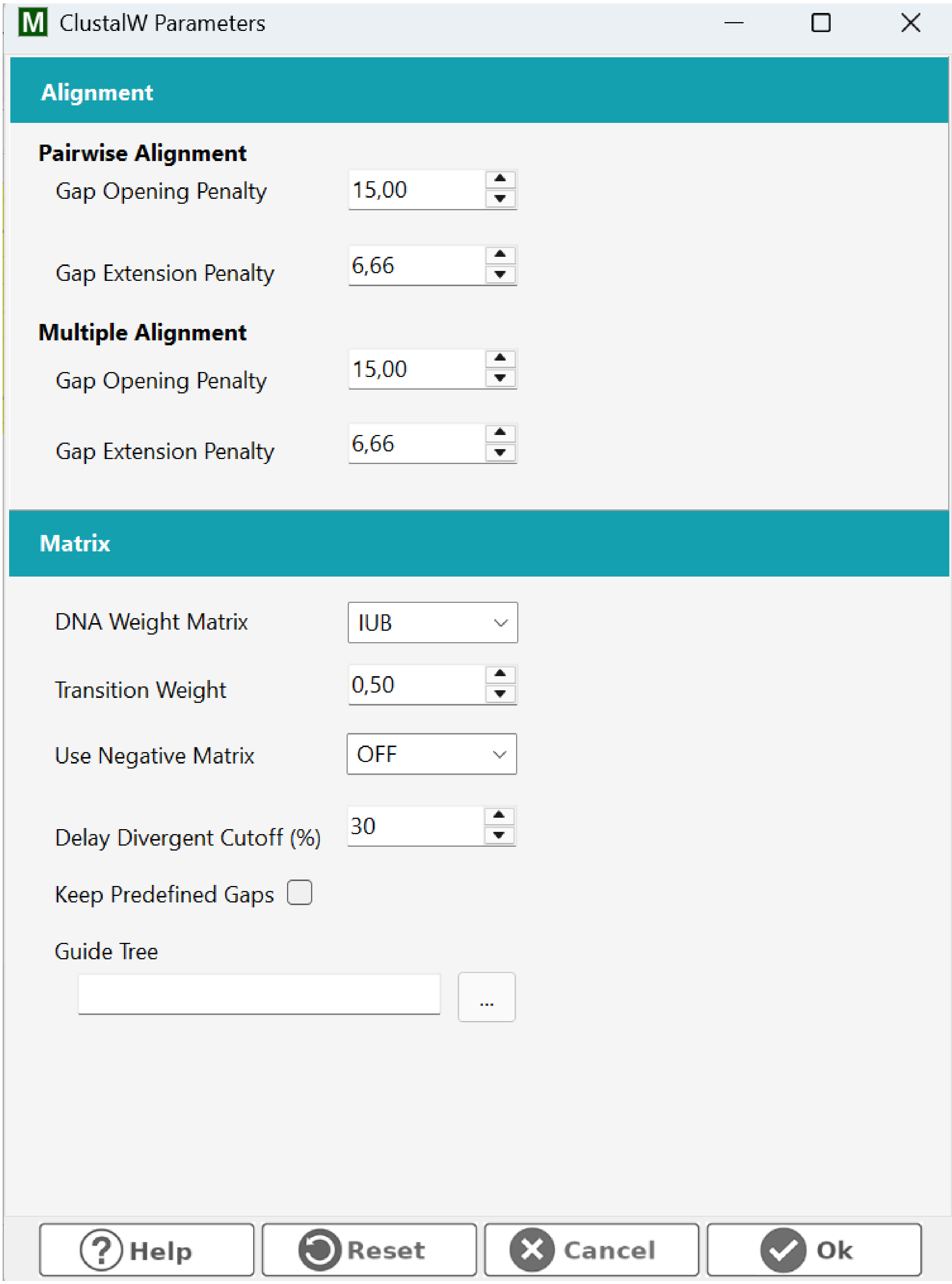
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Pairwise Alignment
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Multiple Alignment
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Gap Extension Penalty
6,66
Matrix
DNA Weight Matrix
IUB
Transition Weight
0,50
Use Negative Matrix
OFF
Delay Divergent Cutoff (%)
30
Keep Predefined Gaps
Guide Tree
...
Help
Reset
Cancel
Ok

M12: Analysis Preferences

Distance Estimation

| Option | Setting |
|---|---|
| **ANALYSIS** | |
| Scope | *Pairs of taxa* |
| **ESTIMATE VARIANCE** | |
| Variance Estimation Method | *Bootstrap method* |
| Bootstrap Replicates | *3000* |
| **SUBSTITUTION MODEL** | |
| Substitutions Type | *Nucleotide* |
| Model/Method | *Maximum Composite Likelihood* |
| Substitutions to Include | *d: Transitions + Transversions* |
| **RATES AND PATTERNS** | |
| Rates among Sites | *Uniform Rates* |
| Pattern among Lineages | *Same (Homogeneous)* |
| **DATA SUBSET TO USE** | |
| Gaps/Missing Data | *Pairwise deletion* |
| Select Codon Positions | ☑ 1st ☑ 2nd ☑ 3rd ☑ Noncoding Sites |

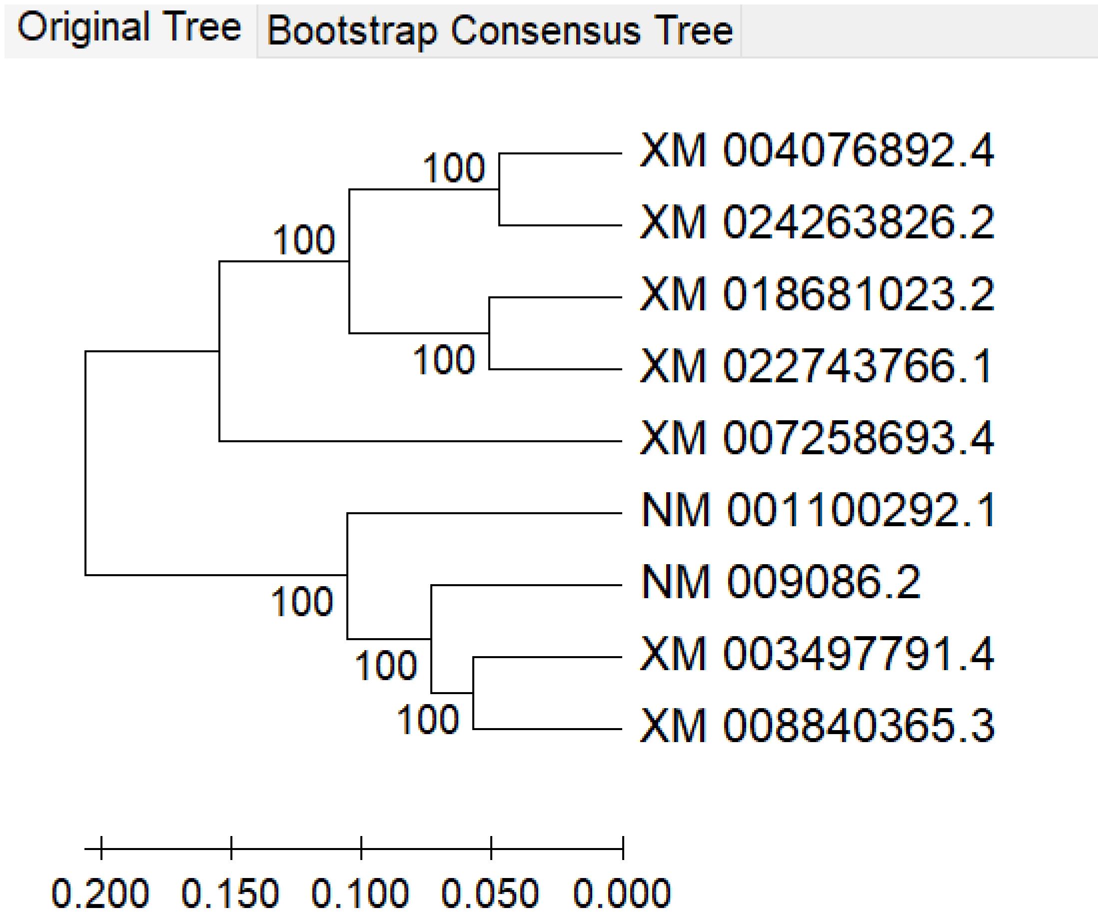
Help Cancel Ok

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1. NM 009086.2 | | 0.0089 | 0.0065 | 0.0144 | 0.0139 | 0.0070 | 0.0140 | 0.0136 | 0.0131 |
| 2. NM 001100292.1 | 0.2448 | | 0.0087 | 0.0145 | 0.0136 | 0.0072 | 0.0133 | 0.0132 | 0.0127 |
| 3. XM 003497791.4 | 0.1433 | 0.2284 | | 0.0143 | 0.0138 | 0.0060 | 0.0142 | 0.0138 | 0.0132 |
| 4. XM 004076892.4 | 0.4683 | 0.4708 | 0.4577 | | 0.0123 | 0.0136 | 0.0091 | 0.0087 | 0.0055 |
| 5. XM 007258693.4 | 0.4206 | 0.4222 | 0.4176 | 0.3568 | | 0.0135 | 0.0107 | 0.0109 | 0.0111 |
| 6. XM 008840365.3 | 0.1502 | 0.1586 | 0.1144 | 0.4053 | 0.3957 | | 0.0139 | 0.0135 | 0.0133 |
| 7. XM 018681023.2 | 0.4264 | 0.4015 | 0.4204 | 0.2364 | 0.2926 | 0.4091 | | 0.0056 | 0.0084 |
| 8. XM 022743766.1 | 0.4137 | 0.4080 | 0.4168 | 0.2282 | 0.3021 | 0.3954 | 0.1019 | | 0.0081 |
| 9. XM 024263826.2 | 0.3802 | 0.3648 | 0.3739 | 0.0939 | 0.2894 | 0.3852 | 0.1986 | 0.1762 | |

M12: Analysis Preferences

Phylogeny Reconstruction

| Option | Setting |
|---|---|
| **ANALYSIS** | |
| Scope | *All Selected Taxa* |
| Statistical Method | *UPGMA* |
| **PHYLOGENY TEST** | |
| Test of Phylogeny | *Bootstrap method* |
| Bootstrap Replicates | *3000* |
| **SUBSTITUTION MODEL** | |
| Substitutions Type | *Nucleotide* |
| Model/Method | *Maximum Composite Likelihood* |
| Substitutions to Include | *d: Transitions + Transversions* |
| **RATES AND PATTERNS** | |
| Rates among Sites | *Uniform Rates* |
| Pattern among Lineages | *Same (Homogeneous)* |
| **DATA SUBSET TO USE** | |
| Gaps/Missing Data | *Pairwise deletion* |
| Select Codon Positions | [x] 1st [x] 2nd [x] 3rd [x] Noncoding Sites |
| **SYSTEM RESOURCE USAGE** | |
| Number of Threads | *3* |


Original Tree
Bootstrap Consensus Tree
100
100
100
100
100
100
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XM 024263826.2
XM 018681023.2
XM 022743766.1
XM 007258693.4
NM 001100292.1
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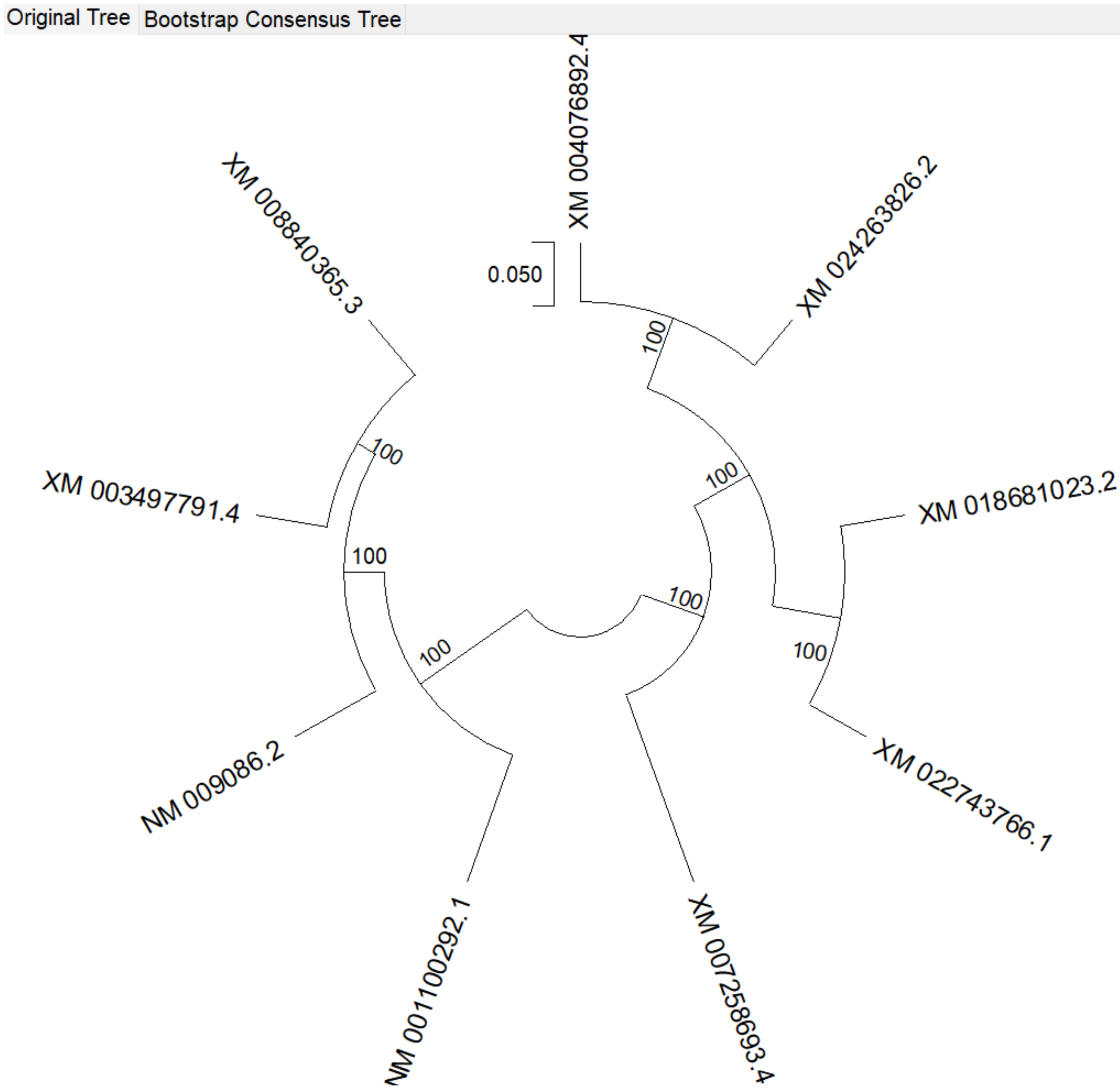


Дерево демонстрирует чёткую дивергенцию между группами, что может отражать различия в их эволюционной истории или адаптации.

Группировка последовательностей XM и NM свидетельствует о возможной функциональной или эволюционной близости между этими генами.